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SHOCK DETECTION AND MANAGEMENT SYSTEM

Abstract

A clinical care system that integrates and analyzes patient data and applies rules to recommend potential treatments. A potential application is detection and management of shock, using hemodynamic data collected from devices such as a right heart catheter. The system may calculate confidence levels for each rule and present high-ranked treatment options to clinicians along with their confidence levels. Confidence levels for treatments may change continuously as a patient's condition evolves. For shock, features extracted from measured data may include for example a cardiac index (cardiac output divided by patient body surface area), systemic vascular resistance, and mean arterial pressure; treatment recommendations derived from these (and other) features may include administration of various medications such as epinephrine and vasopressin, installation of a ventricular assist device, transfusion, and volume resuscitation. Machine learning may be used to match recommendations to current clinical practice, or to optimize patient outcomes.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] One or more embodiments of the invention are related to the field of health care information systems and medical devices. More particularly, but not by way of limitation, one or more embodiments of the invention enable a shock detection and management system.

Description of the Related Art

[0002] Patients in critical condition (or at risk of developing critical complications) must be closely monitored and appropriate treatments must be provided and adjusted quickly to prevent deterioration. In many environments, clinicians must rely on partial information to determine a patient's condition, and they must make care decisions without consulting references or guidelines because of the urgency of the situation. This situation has contributed to relatively high levels of mortality and morbidity in critical care environments.

[0003] Patients experiencing or at risk of developing shock present particular challenges for clinicians. Shock is a life-threatening condition that occurs when the body's organs and tissues do not receive an adequate supply of blood. Insufficient perfusion can lead to organ damage, physiological decompensation and, without medical intervention, can be fatal. There are several types of shock, including hypovolemic (due to severe blood or fluid loss), septic (due to infection), anaphylactic (due to allergic reaction), and cardiogenic, among others. Cardiogenic shock occurs when the heart is unable to pump enough blood to the body due to an ischemic heart attack, heart disease, or a condition that directly affects the heart's ability to function, such as cardiomyopathy, heart failure or arrhythmia. When the heart fails to pump blood efficiently, the other organs in the body do not receive the oxygen and nutrients they need to function properly, which can lead to shock. Consequently, shock is an acute condition that requires immediate medical attention to avoid long-term complications and reduce the potential for death.

[0004] Often, patients with cardiogenic shock do not receive right-heart catheterization (or pulmonary artery catheterization) which provides critical diagnostic information about the hemodynamic state (pressures and blood flow within the heart) of the patient. Such information is essential to determining the cause and severity of the shock and monitoring the patient's response to treatment.

[0005] One reason for the apparent undertreatment is the lack of decision support for associating the complex hemodynamic measurements with actionable information. In addition to the lack of fundamental diagnostic measurements, clinicians treating shock often must manually collect a variety of measurements and perform calculations prior to making decisions. For example, the common measurements used to currently manage shock patients involves the use of information derived from intravenous fluid and drug administration, laboratory measurements, oxygenation and/or mechanical ventilation, vasopressor support, bedside ECG and vital sign monitoring,

hemodynamic monitoring, and mechanical circulatory support devices. This information may be used to determine risk scores, such as FICK and PAPI, and for other fundamental calculations, such as pulmonary vascular resistance, through time.

[0006] Consequently, clinicians must manage a variety of measurements and data points that vary through time, perform manual averaging and calculations and then, from a wide variety of evidence-based publications, determine the optimal treatment for the patient multiple times per hour.

[0007] The urgency of this process is underscored by the patient's acute situation and the deterioration that is accelerated by the body's attempt to compensate for inadequate blood flow. For example, the decreased cardiac output of the diseased heart may activate the sympathetic nervous system leading to the release of adrenaline and noradrenaline which redirect blood to vital organs, increases heart rate and increases the strength of each heart contraction. These mechanisms and others increase the strain on the compromised heart by increasing the resistance against which the heart must pump as well as the amount of blood returning to the heart. The net result of these compensatory actions is a further decrease in cardiac output and an increase in lactic acid build-up (and metabolic acidosis) due to (tissue) hypoxemia.

[0008] Consequently, while the clinician is burdened with interpreting the time series of dynamic patient measurements and selecting a supportive intervention, the patient may experience a downward spiral that requires an immediate diagnosis and course of treatment. This challenging situation has likely contributed to the high (27%-51%) in-hospital mortality rate.

[0009] Although the medical literature includes various protocols and procedures that describes how to treat the various manifestations of shock as well as how to identify them, there are no known systems that integrate these guidelines into an automated clinical decision support system. There are no known systems that collect and integrate comprehensive data on the condition of a patient at risk for shock, and that analyze this data to provide timely treatment recommendations to clinicians.

[0010] For at least the limitations described above there is a need for a shock detection and management system.

BRIEF SUMMARY OF THE INVENTION

[0011] One or more embodiments of the invention may enable a shock detection and management system. The system may provide clinical decision support by collecting and analyzing patient data from multiple sensors or other data sources, and by recommending treatment options based on the analysis.

[0012] One or more embodiments of the invention may include a processor that is coupled to a display that is viewable by one or more clinicians that provide care to a patient at risk for shock. The processor may also be coupled to one or more devices that measure or record multiple clinical parameters associated with the patient's physiological status. It may also be coupled to a memory that contains multiple features, each of which is selected from or derived from the clinical parameters. The features may include one or both of cardiac output, and a cardiac index that is the cardiac output divided by the patient's body surface area. The features may also include mean arterial pressure, and systemic vascular resistance that is the cardiac output divided by the mean arterial pressure. The memory may also include several rules, where each rule includes: a treatment recommendation, one or more activation functions that each map a value of a feature into an activation function value, a weight associated with each activation function, and a confidence function that maps feature values into a confidence level that the treatment recommendation is beneficial for the patient. The confidence function may be calculated by applying an aggregation function to the activation function values using the weight associated with each activation function. The processor may be configured to: obtain values of the clinical parameters from the devices, calculate values of the features from the values of the clinical parameters, calculate the confidence level for each rule using the rule's confidence function, select two or more recommended rules that

have the highest confidence levels, and transmit the recommended rules and their associated confidence levels to the display. The processor may update the recommended rules and confidence levels over time as values of the clinical parameters change over time.

[0013] In one or more embodiments, the patient at risk for shock may be at risk for one or more of cardiogenic shock, hypovolemic shocks, septic shock, and anaphylactic shock.

[0014] In one or more embodiments, the features may further include hemoglobin level in blood, mixed venous oxygen saturation, pulmonary capillary wedge pressure, heart rate, and pulmonary vascular resistance.

[0015] In one or more embodiments, the treatment recommendations associated with the rules may include: start administration of dobutamine, start administration of milrinone, start administration of clevidipine, start administration of phenylephrine, start administration of norepinephrine, start administration of vasopressin, and start administration of epinephrine. In one or more embodiments the treatment recommendations may further include: install ventricular assist device, perform transfusion, and perform volume resuscitation.

[0016] In one or more embodiments, the devices may include a right heart catheter and an associated hemodynamic monitor. In one or more embodiments, the clinical parameters measured by the right heart catheter may include the cardiac output, the cardiac index, the systemic vascular resistance, pulmonary catheter wedge pressure, central venous pressure, and pulmonary vascular resistance. In one or more embodiments, the devices may further include one or more of a central venous catheter, a sphygmomanometer, a laboratory information system, an electronic medical record, and a noninvasive cardiac output monitor.

[0017] In one or more embodiments, the devices may also include a ventricular assist device and a ventilator.

[0018] In one or more embodiments, the processor may also be coupled to a user interface via which the clinicians can accept or reject one or more of the recommended rules. In one or more embodiments, the clinicians may also be able to enter notes via the user interface that explain their acceptance or rejection of one or more of the recommended rules. In one or more embodiments, the processor may also be configured to perform an analysis of the notes and the acceptance or rejection of one or more of the recommended rules, and to modify the rules based on this analysis.

[0019] In one or more embodiments, the aggregation function may be a weighted average using the weight associated with each activation function and with each rule.

[0020] In one or more embodiments, each activation function may be a monotonically nondecreasing or monotonically nonincreasing function of the feature associated with the activation function. In one or more embodiments, each activation function may be a piecewise linear function.

[0021] In one or more embodiments, the processor may also be configured to generate one or more plots of the physiological status of the patient based on values of the features, and to transmit these plots to the display. In one or more embodiments, the plots of physiological status may include a two-dimensional plot of the value of the cardiac index on one axis, and the value of the systemic vascular resistance on the other axis.

[0022] In one or more embodiments, the system may also include a machine learning system coupled to the processor. The machine learning system may be configured to receive values of the features for multiple patients at risk for shock, receive data on the treatments performed by clinicians on these patients, generate a training dataset with samples having the feature values as inputs and the treatments performed as outputs, train a supervised learning model using the training dataset, and generate the confidence function associated with each rule based on the supervised learning model.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The above and other aspects, features and advantages of the invention will be more apparent from the following more particular description thereof, presented in conjunction with the following drawings wherein:

[0024] FIG. 1 illustrates a typical process used in the prior art to manage patients at risk for shock: clinicians receive patient information from a number of different sources, and empirically make care decisions based on training and guidelines that may be incomplete or conflicting.

[0025] FIG. 2 shows an overview architectural diagram of an embodiment of the invention, which integrates patient data from multiple devices and sources, and applies rules to this data to calculate a set of treatments with the highest degree of confidence for the patient.

[0026] FIG. 3 shows an illustrative method of calculating the confidence level for each rule: features are selected from or calculated from measured clinical parameters; the features are input into a set of activation functions associated with each rule, and the activation scores are aggregated into a confidence score for each rule.

[0027] FIGS. 4A and 4B contrast a traditional rule guidelines approach based on sharp criteria for when a rule applies (shown in FIG. 4A), with the graduated rule confidence methodology of one or more embodiments of the invention (shown in FIG. 4B).

[0028] FIG. 5A shows illustrative devices and data sources that may be used to measure patient clinical parameters, and illustrative features that may be selected or derived from these clinical parameters to drive the rule confidence calculations.

[0029] FIG. 5B shows illustrative treatment recommendations that may be associated with a set of rules in one or more embodiments of the invention.

[0030] FIG. 6A shows a set of rules in an illustrative embodiment of the invention that uses only three features: cardiac index (CI), systemic vascular resistance (SVR), and mean arterial pressure (MAP).

[0031] FIG. 6B shows an expanded set of rules in an illustrative embodiment of the invention that uses 9 features to calculate confidence levels for 12 rules.

[0032] FIG. 7 shows illustrative displays of patient state using a two-dimensional grid of the CI and SVR values, and a one-dimensional grid of volume measured by CVP and/or PCWP; these state displays may be shown along with treatment recommendations and their calculated confidence levels.

[0033] FIG. 8 illustrates how rules may be cross-referenced to the patient states in which the rules apply.

[0034] FIG. 9 illustrates an embodiment of the invention in which clinicians may accept or reject any of the suggested treatments.

[0035] FIG. 10 shows an illustrative embodiment of the invention that derives rule confidence functions using machine learning, with a training dataset that may be derived for example from electronic medical records, expert judgements, and clinician treatment selections.

DETAILED DESCRIPTION OF THE INVENTION

[0036] A shock detection and management system will now be described. In the following exemplary description, numerous specific details are set forth in order to provide a more thorough understanding of embodiments of the invention. It will be apparent, however, to an artisan of ordinary skill that the present invention may be practiced without incorporating all aspects of the specific details described herein. In other instances, specific features, quantities, or measurements well known to those of ordinary skill in the art have not been described in detail so as not to obscure the invention. Readers should note that although examples of the invention are set forth herein, the claims, and the full scope of any equivalents, are what define the metes and bounds of the invention.

[0037] For patients in critical condition (or at risk of developing critical complications), selecting

and applying appropriate treatments is both complex and time critical. An important example is patients in cardiac care units who are experiencing or are at risk for shock (cardiogenic, hypovolemic, septic, or anaphylactic for example). Timely and effective treatment is essential to stabilize these patients. However, the optimal treatment and optimal timing depend on many factors. In a typical care setting, many of these factors are either unmeasured or are measured by disparate systems that are not integrated. FIG. 1 illustrates this situation. Patient **101**, who may be for example in a cardiac care unit, is at risk for shock. The patient is currently monitored by device **103** that provides information to display **104**. Although basic information such as heart rate and blood pressure may be available in real-time, many of the complex hemodynamic factors of the patient's condition are not directly measured. Clinician **102** caring for the patient may also receive, for example, results from laboratory tests **105**. Based on this relatively limited information, clinician **102** must make treatment decisions quickly and adjust treatments as the patient's condition evolves. These decisions may be based for example on any of a large number of references **106** that provide recommendations for care of cardiac patients. However, clinician **102** likely has no time to consult the references and must rely on prior training and simple rules of thumb. Moreover, references **106** may provide many different rules such as **107a**, **107b**, **107c**, **d**, etc., which may be contradictory in some situations. It is not feasible for the clinician to remember or to integrate all of these rules manually into a coherent care plan in real time.

[0038] One or more embodiments of the invention may address the issues illustrated in FIG. 1 by integrating patient data from multiple sources and by applying automated rules to the patient data to generate treatment recommendations for the clinician that are updated as the patient condition changes. FIG. 2 shows an overview of an illustrative system **200** with a processor or processors **210** that receive and process data from one or more devices or other data sources that measure the condition of patient **101**. Processor(s) **210** may be for example, without limitation, a server, a desktop computer, a laptop computer, a tablet, a smartphone, a CPU, a GPU, or a network of any of these devices. Processor **210** may be coupled to devices and data sources with any type or types of links or network connections. The illustrative embodiment in FIG. 2 shows four devices **201**, **202**, **203**, and **204** coupled to patient **101**. These devices may measure any types of clinical or physiological parameters of the patient. Some devices may not be directly coupled to the patient but may provide other data on the patient's status or condition; these devices may include for example, without limitation, an electronic medical record, or a laboratory system. For embodiments that provide clinical decision support for cardiac patients, the devices may include a right heart catheter **204** that is configured to measure hemodynamic parameters such as cardiac output. In one or more embodiments, system **200** may manage data from multiple patients simultaneously, potentially in multiple locations.

[0039] Processor **210** may be coupled to a memory that contains a database **211** of rules. Each rule in database **211** has an associated treatment recommendation and information that describes when or to what extent this treatment recommendation is applicable. Database **211** may be organized in any manner (such as SQL or non-SQL databases, files of any format, one or more object stores, or in-memory data structures) and may include code as well as data. Database **211** may include any number of rules. Using rules information **211**, processor **210** performs calculations **212** to obtain a confidence level for each of the rules that may apply to patient **101**. Output **213** from calculations **212** may include for example a table **213** that describes the treatment recommendation **214** associated with each rule, and the calculated confidence level **215** for the rule given the patient's current state. The confidence level provides a relative measure of the likelihood that the associated treatment would be beneficial or appropriate for patient **101** based on the current or most recently measured parameters obtained from the devices.

[0040] After making confidence level calculations **212**, processor **210** may then perform step **216** to select a subset of the rules with the highest confidence levels for the current patient state. These top-ranked rules or associated rule descriptions may then be transmitted to a display **217** coupled to

the processor and viewable by clinician **102** caring for patient **101**. A set of recommendations **218** associated with the top-ranked rules may be displayed along with the calculated confidence levels. A benefit of this system is that multiple recommendations may be displayed along with confidence levels, allowing clinician **102** to select from the top-ranked rules; this approach allows clinicians to apply their own judgement to select an optimal treatment from the options presented, rather than simply presenting a single recommendation with unwarranted certainty.

[0041] Confidence level calculations **212** may be performed in any manner, using any of the data received from devices and using any algorithms. FIG. 3 illustrates one approach to calculating confidence levels that may be used in one or more embodiments of the invention. In this approach, values of clinical parameters **301** are collected from devices such as **201**, **202**, **203**, etc., and step **302** calculates a set of features such as **303**, **304**, **305**, and **306** from the clinical parameters **301**. Calculations **302** may for example perform any data cleaning steps or transformations on the raw clinical parameters **301**, such as smoothing, rescaling, filling missing data, and rejection of outliers or implausible values. They may also derive features by combining clinical parameters into features, for example by deriving ratios, sums, linear or nonlinear combinations, minimums, or maximums, etc. Some clinical parameters may be selected directly as features.

[0042] Subsequently, separate calculations are performed for each rule to determine the rule's confidence level for the patient's current state. Each rule has an associated treatment recommendation; for example, rule **311** has treatment recommendation **312**, and rule **313** has treatment recommendation **314**. FIG. 3 illustrates the confidence level calculation for rule **311**. Feature values may first be input into activation functions that may reflect whether and to what extent the value of each feature implies that the rule should be activated, and the associated treatment should be recommended. Each rule may have any number of associated activation functions. In this illustrative example, for rule **311** activation functions **321**, **322**, **323**, and **324** each map the value of an associated feature into a value between 0 and 1, where a 0 indicates that the feature has no effect on the rule confidence, and a 1 indicates that the feature has its maximum effect on the rule confidence. The activation functions illustrated are all monotonically nondecreasing (functions **321** and **322**) or monotonically nonincreasing (functions **323** and **324**). For illustration, activation functions **321**, **322**, and **323** are piecewise linear, and activation function **324** is a logistic function. Any type of activation function may be used in one or more embodiments of the invention. Some features may be input into more than one activation function; for example, feature **305** is input into activation functions **322** and **323**. Some features may not apply to one or more rules, in which case the feature value may not be input into any activation function associated with the rule; for example, feature **304** is not used in rule **311**.

[0043] The values of activation functions **321**, **322**, **323**, and **324** may then be aggregated in step **330** to calculate the confidence level **215** for the associated rule. Aggregation may use a weight associated with each activation function output, to reflect that activation function's relative importance in determining the rule's confidence level **215**. One or more embodiments of the invention may use any types of aggregation functions, and different rules may have different types of aggregation functions. Aggregation functions may include for example, weighted sums or averages, weighted products or geometric means, sums of products or products of sums, or minimums or maximums. For example, aggregation function **330** for rule **311** may be a weighted average **330a** that multiplies each activation function output by the associated weight, sums these products, and divides the result by the sum of the weights. This illustrative formula **330a** implies that the confidence level **215** will be 1 when all activation functions output 1 and will be 0 when all activation functions output 0.

[0044] The rule confidence level calculation illustrated in FIG. 3 implies that that the confidence level for a rule will change gradually and continuously as values of the features change over time. This approach is substantially different from traditional clinical guidelines, which typically recommend a specific treatment only when features are within a specific range. FIGS. 4A and 4B

illustrate the difference between the classical rule approach (shown in FIG. 4A), and the graduated confidence level approach used in one or more embodiments of the invention (shown in FIG. 4B). In FIG. 4A, a classical rule **401** for treatment of cardiac patients is applied to two features: MAP (mean arterial pressure) and SVR (systemic vascular resistance). This rule is a binary on/off rule with sharp activation boundaries. Graphs **402** of MAP over time (on the left vertical axis) and **403** of SVR over time (on the right vertical axis) show the evolution of an illustrative patient whose condition may be deteriorating. Given classical rule **401**, the rule is active when SVR **403** falls below threshold value **413** and MAP **402** falls below threshold value **412**. Graph **415** shows the corresponding activation of rule **401**, with a 1 value indicating that the rule is active, and 0 indicating that the rule is not active. In this example, rule **401** is briefly activated during period **416**, and then becomes inactive during period **417**, only to become active again in period **418**. This on/off behavior does not reflect the continuous change in the patient's condition and may mislead clinicians more than it assists them.

[0045] In contrast, FIG. 4B shows a graduated confidence level approach **421** for rule **401**, with the same patient evolution **402** and **403**. An activation function **422** is applied to MAP value **402**, and activation function **423** is applied to SVR value **403**; the activation function outputs are combined in a weighted average using weights 0.6 for MAP and 0.4 for SVR. (These specific activation functions and weights are illustrative.) The resulting rule confidence value **425** changes continuously as the patient condition evolves, increasing gradually from 0 at the beginning of the time interval to a value near 1.0 at the end of the time interval. During period **417** (in FIG. 4A) when the classical rule is deactivated since the MAP value fluctuates briefly above threshold **412**, the confidence level for graduated rule **421** dips slightly to value **426**, but it does not go immediately to 0. This continuous change in rule confidence reflects the evolving patient condition more directly than the on/off behavior of the classical rule and may be more intuitive for a clinician to evaluate and use.

[0046] The framework described above for generating confidence levels for a set of rules based on patient data may be applied to any types of patient conditions. FIGS. 5A, 5B, 6A, and 6B describe specific embodiments that may be applied for example to cardiac patients that are at risk for shock. FIG. 5A shows illustrative devices **501** that may be used to collect patient data. These devices may include for example a Swan-Ganz catheter **204** with an associated hemodynamic monitor that collects various parameters of heart function. They may also include systems not directly coupled to the patient, such as an electronic medical record (EMR) system **502**, and a laboratory system **503** that may for example analyze the patient's blood for hemoglobin level or other characteristics. Other devices may include for example, without limitation, any type of heart catheter or central venous catheter, a Doppler ultrasound, an echocardiogram, a sphygmomanometer, a pulse rate monitor, and a respiratory rate monitor. One or more embodiments may use any combinations or subsets of these devices, or any additional devices as needed.

[0047] FIG. 5A also shows illustrative features **511** that may be selected from or derived from the clinical parameters measured by devices **501**. Cardiac Output (CO) **512** may be defined for example as the amount of blood pumped per minute, which may be measured for example by a right heart catheter (or other devices). Cardiac Index (CI) **513** may be defined for example as CO **512** divided by the patient's body surface area **514**. Body Surface Area **514** may be estimated for example using the formula: $\text{Body Surface Area} = (0.007184) * (\text{Height}(\text{cm})^{\frac{1}{2}} * 0.725) * (\text{Weight}(\text{kg})^{\frac{1}{2}} * 0.425)$. Mean Arterial Pressure (MAP) **515** may be calculated for example as a weighted average of the systolic and diastolic pressure, using for example the formula: $\text{MAP} = (\frac{1}{3}) * \text{SP} + (\frac{2}{3}) * \text{DP}$. Systemic Vascular Resistance (SVR) **516** may be calculated for example as CO **512** divided by MAP **515**. The three features **513**, **515**, and **516** may be used in one or more embodiments as a minimal set of features to drive calculations of confidence levels for certain rules, as described below with respect to FIG. 6A. Other illustrative features that may be used in one or more embodiments of the invention may include, for example,

any or all of Hemoglobin Level (Hgb), Mixed Venous Oxygenation Level (SvO₂), Pulmonary Capillary Wedge Pressure (PCWP), Pulmonary Vascular Resistance (PVR), Central Venous Pressure (CVP), Respiratory Rate (RR), and Heart Rate (HR).

[0048] The table below provides a summary of features **511**:

TABLE-US-00001 Symbol Description Measured By Formula CO Cardiac Output: the amount of Right Heart Catheter, in the Stroke volume blood your heart pumps per right pulmonary artery. x Heart Rate minute. Doppler Ultrasound: Uses an ultrasound machine with a special probe that measures the Doppler shift in the returning ultrasound waves to decipher the blood flow rate and volume, both of which lead to the cardiac index. Echocardiogram: Uses two-dimensional ultrasound paired with Doppler shift measurements to elucidate blood flow rate and volume. Modified carbon dioxide Fick method: Utilizes the Fick principle and measures changes in CO₂ elimination and end-tidal CO₂ (which is a measure of atrial CO₂). Hgb Hemoglobin: protein that Hemoglobin blood test, carries oxygen and carbon number of hemoglobin dioxides in blood present in red blood cells. SvO₂ Indicates the level of Swan-Ganz Catheter, central oxygenation of mixed venous venous cannulation of the blood returning to the heart superior vena cava or right from the body atrium PCWP Pulmonary Capillary Wedge Swan-Ganz catheter, central Pressure: used to assess left vein and advancing the ventricular filling, represent left catheter into a branch of the atrial pressure, and assess mitral pulmonary artery valve function. CI Cardiac Index: Turns cardiac CO/Body output into a normalized value Surface Area that accounts for the body size of the patient CVP Central Venous Pressure: Measured by a central Measure of pressure in the vena venous catheter placed cava, can be used as an through either the estimation of preload and right subclavian or internal atrial pressure jugular veins. MAP Mean Arterial Pressure: Invasive arterial catheter $MAP = \text{Average arterial pressure and non-invasive } \frac{2}{3} * DP + \text{throughout one cardiac cycle, intermittent } \frac{1}{3} * SP$ systole, and diastole. sphygmomanometer are the standard ways to measure both systolic and diastolic blood pressures. Once these values are known, a MAP value can easily be determined. An oscillometric blood pressure device can also be used to measure MAP. RR Respiratory Rate: The number Different technologies are of breaths per minute, is highly available for measuring. In regulated to enable cells to contact-based measuring produce the optimum amount of techniques, the sensor (i.e., energy at any given occasion the element directly affected by the measurand) must be in contact with the subject's body. HR Heart Rate: The number of Where the pulse is palpated beats per minute. The intrinsic on the radial aspect of the rate of the SA node is typically forearm, just proximal to the around 60 to 100 beats per wrist joint. minute (BPM). SVR Systemic Vascular Resistance, Right Heart Catheterization. $SVR = \text{or, Total peripheral resistance}$ SVR may be estimated if CO/MAP (TPR), is the amount of force one can get an accurate exerted on circulating blood by blood pressure reading and the vasculature of the body. the patient's cardiac output, which can be estimated using ultrasound data. The BP can be used to calculate the MAP, and this can be plugged into the above equation to calculate SVR. PVR Pulmonary Vascular This measurement is Resistance: resistance against obtained through a right blood flow from the pulmonary heart catheterization (e.g., artery to the left atrium. Swan-Ganz catheters).

[0049] FIG. 5B shows illustrative treatment recommendations **531** that may be generated in one or more embodiments of the invention via confidence level calculations **521** based on values of features **511**. As described above, in one or more embodiments the confidence level **521** for each treatment recommendation may be calculated using activation functions **522** associated with features **511**, whose outputs are combined using weights **523** and an aggregation function **524**. Treatment recommendations **532** each recommend administration of various medications, including for example dobutamine, milrinone, clevidipine, phenylephrine, norepinephrine, vasopressin, and epinephrine. In some embodiments a rule may have a treatment recommendation for a specific quantity of an associated medication. Other treatment recommendations associated with rules in one or more embodiments may include for example, without limitation, installation of a ventricular

assist device (VAD) **533**, performing a transfusion **534**, and performing volume resuscitation (**535**). [0050] FIGS. **6A** and **6B** show two different sets of rules that may be used in one or more embodiments to manage the risk of shock. FIG. **6A** shows a relatively small set of rules **601** that are based on the values of only three features: Cardiac Index **513**, Systemic Vascular Resistance **516**, and Mean Arterial Pressure **515**. Each rule has an associated treatment recommendation **532**. Table **601** shows the ranges of each feature within which each rule has an associated activation function that exceeds a threshold value, such as 0.60 for example; features without ranges associated with a rule do not have an associated activation function for that rule. For example, rule **602** may have activation functions **423** for SVR and **422** for MAP as shown in FIG. **4B**. FIG. **6B** shows a more extensive set of illustrative rules **611** that depend upon the values of 9 different features; these rules also incorporate a wider range of treatment recommendations **612**. As in FIG. **6A**, the ranges shown for features may correspond to feature values with corresponding activation function values greater than a threshold such as 0.60.

[0051] The specific rules, treatment recommendations, features, and activation ranges shown in FIGS. **6A** and **6B** are illustrative. One or more embodiments of the invention may use different combinations of features, different ranges, different activation functions and weights, and different treatment recommendations.

[0052] In one or more embodiments of the invention, the system may generate plots or other displays of the patient's physiological status, in addition to calculating rule confidence levels and displaying the top-ranked treatment recommendations. Selected feature values may be plotted for example on one-dimensional or two-dimensional charts or grids. These plots may aid the clinician in understanding the patient's current condition and trajectory, and thereby in understanding why certain treatments are recommended. FIG. **7** shows an illustrative example for patients at risk of shock. As described above, values of clinical parameters **701** are measured by devices, values of features **702** are derived from these clinical parameters, and confidence levels calculated from feature values are used to select a set of applicable rules **703** with treatment recommendations. Feature values **702** may also be used to generate one or more plots of the patient's physiological status. In this example, plot **711** is a two-dimensional display with the Cardiac Index (which measures cardiac output) on one axis, and the Systemic Vascular Resistance on the other axis. Point **712** shows the patient's current state, and trajectory **713** shows the change in the patient's state over time. The patient's volume status, measured by Central Venous Pressure and or Pulmonary Capillary Wedge Pressure, is shown in plot **721**; point **722** shows the current value, and trajectory **723** shows the change over time. These plots **711** and **721** may be shown on display **217**, along with the recommended treatments **218** and their associated confidence levels.

[0053] Because the patient status plots and the rule confidence level calculations are both based on the current values of the features, the rule confidence levels are correlated with the patient status plots. The 3-by-3 grid **711** and 1-by-3 grid **721** may therefore provide a useful method for organizing which rules are applicable in which patient states. This methodology may also help clinicians understand why certain rules are recommended with high confidence in certain states. FIG. **8** shows a subset **611a** of the rules **611** of FIG. **6B**, along with a plot **801** for each rule that shows which portions of the grid **711** each rule is applicable (with a high confidence level). A similar scheme may be used for grid **721**, or for any other type of plot of patient status. For example, rule **802** is applicable with high confidence in the upper left grid square **812**, where Cardiac Output is low and Systemic Vascular Resistance is high. Similarly rule **802** is applicable in the lower right grid squares **813**, where Cardiac Output is moderate or high, and Systemic Vascular Resistance is low.

[0054] In one or more embodiments of the invention, clinicians may have the capability to indicate whether they accept or reject any or all of the recommended treatments, and potentially to explain their reasoning. This information may be used for example to improve the recommendation system over time based on clinician input. FIG. **9** shows an illustrative embodiment with treatment

recommendations **218** shown on display **217**; the display also provides selections **901** for clinician **102** to accept or reject each recommendation. In this example the clinician clicks on or otherwise selects button **902** to indicate that the “start norepinephrine” recommendation is accepted. Clinician **102** may also be able to enter notes **903** explaining this decision. A record of the treatments presented **218**, the choice made **902**, the clinician notes **903**, and the patient's state at the time of the choice may be saved in a database **904** of clinician treatment choices; this information may be collected in database **904** for a set of patients over a period of time. Processor **210** (or any other processor or processors) may perform analysis or analyses **905** of database **904**, and the results of the analysis may be used to modify the clinical decision support system and the rules **211**, for example by adjusting rule confidence calculations to conform more closely to clinician's actual decisions or to improve patient outcomes.

[0055] In one or more embodiments, machine learning techniques may be used to train a system to calculate confidence levels for rules, and to improve these calculations over time. FIG. **10** illustrates an approach to machine learning that uses supervised learning to train a system to match clinician's actual or preferred treatment choices. Training data **1001** may for example consist of a collection **1005** of labelled samples, with feature vectors **1006** that may correspond for example to any or all of the features described above, and with the label **1007** indicating the selected or desired treatment. Sources for training data **1001** may include for example, without limitation, electronic medical records **1002**, clinician treatment choices **904** captured by the clinical decision support system (as described above with respect to FIG. **9**), and expert opinions **1003** on what treatments are best for certain feature vectors. One or more embodiments may use any type of machine learning, including but not limited to neural networks, linear or logistic regression, decision trees, random forests, support vector machines, or nearest neighbors. In the illustrative example shown in FIG. **10**, a neural network **1010** is trained in process **1015** using data **1001**; feature vector values **1006** are input into the input layer **1011** of the network, and the output layer **1012** may be for example a softmax layer that generates confidence levels for each treatment. The confidence levels calculated by the neural network **1010** may be compared in step **1016** to the actual treatments **1007** (for example with a standard loss function), and the network may be iteratively trained using backpropagation.

[0056] Another approach to developing and improving treatment recommendations that may be used in one or more embodiments is to use machine learning techniques to determine treatments that optimize patient outcomes (as opposed to the methodology shown in FIG. **10** that trains the system to match current clinical practice). This approach may require a more extensive training dataset that captures both short-term and long-term patient outcomes, along with the patient status over time and the treatment decisions made. For example, detrimental outcomes may include obvious events such as mortality, as well detrimental exposures (typically exposure to medical interventions such as mechanical ventilation, that increase morbidity and mortality). A cost may be assigned to each type of outcome (for example with higher costs for poorer outcomes), and a system may be trained to perform an optimal action at each point in time to minimize total patient costs.

[0057] While the invention herein disclosed has been described by means of specific embodiments and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

Claims

1. A shock detection and management system comprising: a processor coupled to a display viewable by one or more clinicians that provide care to a patient at risk for shock; one or more devices that measure or record a plurality of clinical parameters associated with a physiological status of the patient; and a memory comprising a plurality of features, each selected from or

derived from the plurality of clinical parameters, wherein the plurality of features comprise one or both of cardiac output; and cardiac index, comprising the cardiac output divided by a body surface area of the patient; mean arterial pressure; and systemic vascular resistance, comprising the cardiac output divided by the mean arterial pressure; and, a multiplicity of rules, wherein each rule comprises a treatment recommendation; one or more activation functions associated with each rule, wherein each activation function of the one or more activation functions maps a value of a feature of the plurality of features into an activation function value; a weight associated with each activation function and with each rule; and a confidence function associated with each rule that maps values of the plurality of features into a confidence level that the treatment recommendation is beneficial for the patient, wherein the confidence function is calculated by applying an aggregation function to activation function values associated with each rule using the weight associated with each activation function and with each rule; wherein the processor is configured to obtain values of the plurality of clinical parameters from the one or more devices; calculate values of the plurality of features from the values of the plurality of clinical parameters; calculate the confidence level for each rule of the multiplicity of rules using the confidence function associated with each rule; select a plurality of recommended rules from the multiplicity of rules that have highest confidence levels; transmit the plurality of recommended rules and their associated confidence levels to the display; and update the plurality of recommended rules and associated confidence levels over time as values of the plurality of clinical parameters change over time.

2. The shock detection and management system of claim 1, wherein the patient at risk for shock is at risk for one or more of cardiogenic shock, hypovolemic shock, septic shock, and anaphylactic shock.

3. The shock detection and management system of claim 1, wherein the plurality of features further comprises: hemoglobin level in blood; mixed venous oxygen saturation; pulmonary capillary wedge pressure; central venous pressure; heart rate; and, pulmonary vascular resistance.

4. The shock detection and management system of claim 1, wherein treatment recommendations associated with the multiplicity of rules comprise: start administration of dobutamine; start administration of milrinone; start administration of clevidipine; start administration of phenylephrine; start administration of norepinephrine; start administration of vasopressin; and, start administration of epinephrine.

5. The shock detection and management system of claim 4, wherein treatment recommendations associated with the multiplicity of rules further comprise: install ventricular assist device; perform transfusion; and, perform volume resuscitation.

6. The shock detection and management system of claim 1, wherein the one or more devices comprise a right heart catheter and an associated hemodynamic monitor.

7. The shock detection and management system of claim 6, wherein clinical parameters measured by the right heart catheter comprise the cardiac output, the cardiac index, the systemic vascular resistance, pulmonary catheter wedge pressure, central venous pressure, and pulmonary vascular resistance.

8. The shock detection and management system of claim 1, wherein the one or more devices further comprise one or more of a central venous catheter; a sphygmomanometer; a laboratory information system; an electronic medical record system; and, a noninvasive cardiac output monitor.

9. The shock detection and management system of claim 1, wherein the one or more devices further comprise one or more of a ventricular assist device; and, a ventilator.

10. The shock detection and management system of claim 1, wherein the processor is further coupled to a user interface via which the one or more clinicians can accept or reject one or more of the plurality of recommended rules.

11. The shock detection and management system of claim 10, wherein the one or more clinicians can further enter notes via the user interface that explain acceptance or rejection of one or more of the plurality of recommended rules.

- 12.** The shock detection and management system of claim 11, wherein the processor is further configured to perform an analysis of the notes and the acceptance or rejection of one or more of the plurality of recommended rules; and modify the multiplicity of rules based on this analysis.
- 13.** The shock detection and management system of claim 1, wherein the aggregation function comprises a weighted average using the weight associated with each activation function and with each rule.
- 14.** The shock detection and management system of claim 1, wherein each activation function is a monotonically nondecreasing or a monotonically nonincreasing function of the feature associated with each activation function.
- 15.** The shock detection and management system of claim 14, wherein each activation function is piecewise linear function.
- 16.** The shock detection and management system of claim 1, wherein the processor is further configured to: generate one or more plots of the physiological status of the patient based on values of the plurality of features; and, transmit the one or more plots to the display.
- 17.** The shock detection and management system of claim 16, wherein the one or more plots of the physiological status of the patient comprise a two-dimensional plot of a value of the cardiac index on one axis and a value of the systemic vascular resistance on a second axis.
- 18.** The shock detection and management system of claim 1, further comprising a machine learning system coupled to the processor and configured to receive values of the plurality of features for a multiplicity of patients at risk for shock; receive data comprising treatments performed by clinicians on the multiplicity of patients; generate a training dataset comprising samples having the values of the plurality of features as inputs and treatments performed as outputs; train a supervised learning model using the training dataset; and, generate the confidence function associated with each rule based on the supervised learning model.
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